## Example 1.9

Find atleast two rational numbers between $\frac{-3}{4}$ and $\frac{-2}{5}$.

***Solution:***

The denominators are different for the given rational numbers. The LCM of the denominators 4 and 5 is 20. Make the rational numbers such that they have common denominators as 20. Here,

$\frac{-3}{4}=\frac{-3}{4}\times\frac{5}{5}=\frac{-15}{20}$ and $\frac{-2}{5}=\frac{-2}{5}\times\frac{4}{4}=\frac{-8}{20}$.

It is easy now to find and insert rational numbers between $\frac{-15}{20}$ and $\frac{-8}{20}$ as shown below.

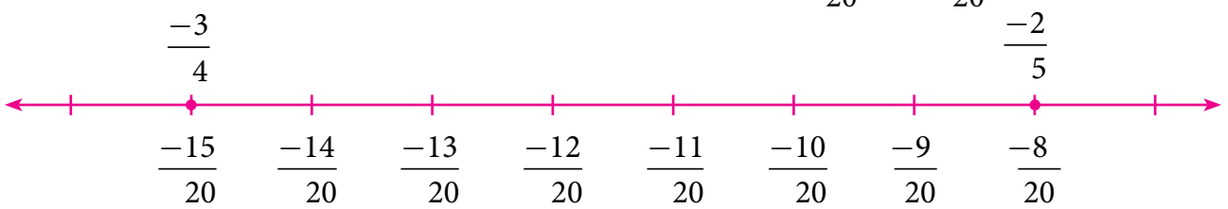



Fig. 1.14

We can list a few rational numbers as $\frac{-14}{20}, \frac{-13}{20}, \frac{-12}{20}, \frac{-11}{20}, \frac{-10}{20}$ and $\frac{-9}{20}$ between $\frac{-15}{20}$ and $\frac{-8}{20}$.

Are these the only rational numbers between $\frac{-15}{20}$ and $\frac{-8}{20}$? Think! Try to find 10 more rational numbers between them, if possible!

### Note

We can find many rational numbers between $\frac{-7}{11}$ and $\frac{5}{-9}$ quickly as given below: The range of rational numbers can be got by the cross multiplication of denominators with the numerators after writing the given fractions in standard form. The cross multiplication here $\frac{-7}{11}\ \frac{-5}{9}$ gives the range of rational numbers from –63 to –55 with the denominator 99. This is nothing but making the given rational numbers equivalent with the denominator 99!

## Exercise 1.1

1. **Fill in the blanks:**

   (i) $\frac{-19}{5}$ lies between the integers _____ and _____.

   (ii) The decimal form of the rational number $\frac{15}{-4}$ is _____.

   (iii) The rational numbers $\frac{-8}{3}$ and $\frac{8}{3}$ are equidistant from _____.

   (iv) The next rational number in the sequence $\frac{-15}{24}, \frac{20}{-32}, \frac{-25}{40}$ is _____.

   (v) The standard form of $\frac{58}{-78}$ is _____.



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